DIAGONALIZATION

Diagonalizable or defective

The best form under similarity depends on whether every eigenvalue has enough eigenvectors.

DIAGONALIZABLE

$$A = PDP^{-1}$$

HOLDS WHEN every eigenvalue has geometric multiplicity equal to its algebraic multiplicity

✓ Diagonalizable

Each eigenvalue contributes 1×1 Jordan blocks

Geometric multiplicity matches algebraic multiplicity: a full set of eigenvectors.

BOUNDARY

✗ Defective: Jordan form

$J_k(\lambda)$: λ on the diagonal, ones on the superdiagonal

A defective eigenvalue contributes blocks larger than 1×1.

$J_2(1) = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$, the horizontal shear

The repeated eigenvalue 1 has just one eigenvector, $(1, 0)$.

A MILLIONTH AWAY

1.000001 Change the corner entry of the shear to 1.000001: the eigenvalues become distinct and the matrix diagonalizes, with eigenvectors $(1, 0)$ and $(1, 0.000001)$ almost parallel.